

## 1. Mean Distance Within a Single Point Group

### Mathematical Formulation

**Inputs:** A single group of  $N$  points, denoted as  $G = \{P_1, P_2, \dots, P_N\}$ . Each point  $P_i$  is defined by its coordinates  $(x_i, y_i)$  in a 2D Cartesian space.

**Point Representation:** Each point  $P_i \in G$  is represented as  $P_i = (x_i, y_i)$ .

**Distance Metric:** The Euclidean distance  $d(P_i, P_j)$  between two points  $P_i = (x_i, y_i)$  and  $P_j = (x_j, y_j)$  is given by:

$$d(P_i, P_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

**Calculation:** The set of all unique pairwise distances within the group  $G$  is  $D_G = \{d(P_i, P_j) \mid 1 \leq i < j \leq N\}$ . The total number of such unique pairs is  $\binom{N}{2} = \frac{N(N-1)}{2}$ .

The mean distance within the point group,  $\mu_G$ , is calculated as:

$$\mu_G = \frac{\sum_{1 \leq i < j \leq N} d(P_i, P_j)}{\binom{N}{2}}$$

For this calculation to be meaningful, the number of points  $N$  must be at least 2.

**Output:** A non-negative scalar value  $\mu_G \in \mathbb{R}_{\geq 0}$ .

## 2. Mean Distance Between Two Distinct Point Groups

### Mathematical Formulation

**Inputs:** A collection of  $M$  points in total,  $P_{\text{total}} = \{P_1, \dots, P_M\}$ , where each point  $P_k = (x_k, y_k)$ . A set of categorical labels  $L = \{l_1, \dots, l_M\}$ , where each  $l_k \in \{0, 1\}$  assigns point  $P_k$  to one of two groups,  $G_0$  or  $G_1$ .

Let  $G_0 = \{P_k \in P_{\text{total}} \mid l_k = 0\}$  be the set of  $N_0$  points in group 0. Let  $G_1 = \{P_k \in P_{\text{total}} \mid l_k = 1\}$  be the set of  $N_1$  points in group 1.

**Distance Metric:** The Euclidean distance  $d(A, B)$  between a point  $A \in G_0$  and a point  $B \in G_1$  is used.

**Calculation:** The set of all distances between points in  $G_0$  and points in  $G_1$  is  $D_{G_0, G_1} = \{d(A, B) \mid A \in G_0, B \in G_1\}$ . The total number of such pairs is  $N_0 \times N_1$ .

The mean distance between groups  $G_0$  and  $G_1$ , denoted  $\mu_{G_0, G_1}$ , is:

$$\mu_{G_0, G_1} = \frac{\sum_{A \in G_0} \sum_{B \in G_1} d(A, B)}{N_0 N_1}$$

This calculation requires both  $N_0 > 0$  and  $N_1 > 0$ .

**Output:** A non-negative scalar value  $\mu_{G_0, G_1} \in \mathbb{R}_{\geq 0}$ .

### 3. Composite Separability and Cohesion Measure

#### Mathematical Formulation

**Inputs:** Same as for the Mean Distance Between Two Distinct Point Groups.

**Component Calculations:**

1. **Mean Inter-Group Distance ( $M_{inter}$ ):**  $M_{inter} = \mu_{G_0, G_1}$ .
2. **Mean Intra-Group Distance for Group 0 ( $M_{intra,0}$ ):** Calculated as  $\mu_{G_0}$  using only points in  $G_0$ . Requires  $N_0 \geq 2$ .
3. **Mean Intra-Group Distance for Group 1 ( $M_{intra,1}$ ):** Calculated as  $\mu_{G_1}$  using only points in  $G_1$ . Requires  $N_1 \geq 2$ .
4. **Average of Mean Intra-Group Distances ( $\bar{M}_{intra}$ ):**

$$\bar{M}_{intra} = \frac{M_{intra,0} + M_{intra,1}}{2}$$

5. **Standard Deviation of Mean Intra-Group Distances ( $\sigma_{M_{intra}}$ ):** Population standard deviation of  $M_{intra,0}$  and  $M_{intra,1}$ :

$$\sigma_{M_{intra}} = \frac{|M_{intra,0} - M_{intra,1}|}{2}$$

**Objective Function Value ( $J$ ):** The objective function  $J$  is defined as:

$$J = M_{inter} + \bar{M}_{intra} - \sigma_{M_{intra}}$$

Substituting the expressions for  $\bar{M}_{intra}$  and  $\sigma_{M_{intra}}$ :

$$J = M_{inter} + \frac{M_{intra,0} + M_{intra,1}}{2} - \frac{|M_{intra,0} - M_{intra,1}|}{2}$$

This expression simplifies to:

$$J = M_{inter} + \min(M_{intra,0}, M_{intra,1})$$

**Output:** A scalar value  $J \in \mathbb{R}$ .